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AEROSOL PROVISION SYSTEM AND METHOD

Abstract

Described is an aerosol provision system comprising a haptic component, an orientation sensor and control circuitry configured. The control circuitry is configured to determine an orientation of the aerosol provision system based on one or more readings from the orientation sensor, and adjust a setting of the haptic component based on the determined orientation.

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to an aerosol provision system and method.

BACKGROUND

[0002] Electronic aerosol provision systems such as electronic cigarettes (e-cigarettes) generally contain an aerosol-generating material, such as a reservoir of a source liquid containing a formulation, typically including nicotine, or a solid material such as a tobacco-based product, from which an aerosol is generated for inhalation by a user, for example through heat vaporisation. Thus, an aerosol provision system will typically comprise an aerosol generator, e.g. a heating element, arranged to aerosolise a portion of aerosol-generating material to generate an aerosol in an aerosol generation region of an air channel through the aerosol provision system. As a user inhales on the device and electrical power is supplied to the aerosol generator, air is drawn into the device through one or more inlet holes and along the air channel to the aerosol generation region, where the air mixes with the vaporised aerosol generator and forms a condensation aerosol. The air drawn through the aerosol generation region continues along the air channel to a mouthpiece, carrying some of the aerosol with it, and out through the mouthpiece for inhalation by the user.

[0003] It is common for aerosol provision systems to comprise a modular assembly, often having two main functional parts, namely an aerosol provision device and an article. Typically, the article will comprise the article aerosol-generating material and the aerosol generator (heating element), while the aerosol provision device part will comprise longer-life items, such as a rechargeable battery, device control circuitry and user interface features. The aerosol provision device may also be referred to as a reusable part or battery section and the article may also be referred to as a consumable, disposable/replaceable part, cartridge or cartomiser.

[0004] The aerosol provision device and article are mechanically coupled together at an interface for use, for example using a screw thread, bayonet, latched or friction fit fixing.

[0005] When the aerosol-generating material in an article has been exhausted, or the user wishes to switch to a different article having a different aerosol-generating material, the article may be removed from the aerosol provision device and a replacement article may be attached to the device in its place.

[0006] Haptic components or elements can form a part of the aerosol provision system and be used to provide a sensory signal or sensation to the user. Determining how and when to provide such a signal such that the user is able to detect and interpret/understand the signal can be challenging.

[0007] Various approaches are described herein which seek to help address or mitigate some of the issues discussed above.

SUMMARY

[0008] The disclosure is defined in the appended claims.

[0009] In accordance with some embodiments described herein, there is provided an aerosol provision system comprising a haptic component, an orientation sensor and control circuitry configured. The control circuitry is configured to determine an orientation of the aerosol provision system based on one or more readings from the orientation sensor, and adjust a setting of the haptic component based on the determined orientation.

[0010] The orientation sensor can be a gyroscope.

[0011] The aerosol provision system can further comprise a movement sensor, wherein the control circuitry is further configured to determine a movement of the aerosol provision system based on one or more readings from the movement sensor, and adjust a setting of the haptic component based on the determined movement. The movement sensor can be an accelerometer. The aerosol provision system can further comprise an Inertial Measurement Unit, IMU, incorporating the orientation sensor and the movement sensor. The aerosol provision system can further comprise a display component, and the control circuitry is further configured to adjust a setting of the display component based on the determined movement. The aerosol provision system can further comprise

a speaker component, and the control circuitry is further configured to adjust a setting of the speaker component based on the determined movement.

[0012] Adjusting the setting of the haptic component can correspond to disabling the haptic component.

[0013] The setting can be an amplitude and/or frequency of a vibration generated by the haptic component.

[0014] The setting can be a magnitude of a force generated by the haptic component.

[0015] The aerosol provision system can further comprise a display component, and the control circuitry is further configured to adjust a setting of the display component based on the determined orientation. The setting can be a brightness of the display component. Adjusting the setting of the display component can correspond to disabling the display component.

[0016] The aerosol provision system can further comprise a speaker component, and the control circuitry is further configured to adjust a setting of the speaker component based on the determined orientation. The setting can be a volume of the speaker component. Adjusting the setting of the speaker component can correspond to disabling the speaker component.

[0017] The aerosol provision system can further comprise a communications interface, and wherein control circuitry is further configured to adjust a setting of the haptic component based on a signal received via the communications interface from an external device.

[0018] The aerosol provision system can further comprise an input device, and wherein control circuitry is further configured to adjust a setting of the haptic component based on an input received via the input device.

[0019] In accordance with some embodiments described herein, there is provided an aerosol provision device for an aerosol provision system, the aerosol provision system comprising a haptic component and an orientation sensor, wherein the aerosol provision device comprises control circuitry configured to: determine an orientation of the aerosol provision system based on one or more readings from the orientation sensor; and adjust a setting of the haptic component based on the determined orientation.

[0020] In accordance with some embodiments described herein, there is provided a method for operating an aerosol provision system comprising determine an orientation of the aerosol provision system based on one or more readings from an orientation sensor of the aerosol provision system; and adjust a setting of a haptic component of the aerosol provision system based on the determined orientation.

[0021] There is also provided a computer readable storage medium comprising instructions which, when executed by a processor, performs the above method.

[0022] These aspects and other aspects will be apparent from the following detailed description. In this regard, particular sections of the description are not to be read in isolation from other sections.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0023] Embodiments of the invention will now be described, by way of example only, with reference to accompanying drawings, in which:

[0024] FIG. 1 is a schematic diagram of an aerosol provision system;

[0025] FIG. 2 is a flow diagram of a method for operating an aerosol provision system.

DETAILED DESCRIPTION

[0026] Aspects and features of certain examples and embodiments are discussed/described herein. Some aspects and features of certain examples and embodiments may be implemented conventionally and these are not discussed/described in detail in the interests of brevity. It will thus be appreciated that aspects and features of articles and systems discussed herein which are not

described in detail may be implemented in accordance with any conventional techniques for implementing such aspects and features.

[0027] The present disclosure relates to aerosol provision systems, which may also be referred to as vapour provision systems, such as e-cigarettes. Throughout the following description the term “e-cigarette” or “electronic cigarette” may sometimes be used, but it will be appreciated this term may be used interchangeably with aerosol provision system and electronic aerosol provision system.

[0028] As noted above, aerosol provision systems (e-cigarettes) often comprise a modular assembly including both a reusable part (aerosol provision device) and a replaceable (disposable) or refillable cartridge part, referred to as an article. Systems conforming to this type of two-part modular configuration may generally be referred to as two-part systems or devices. It is also common for electronic cigarettes to have a generally elongate shape. For the sake of providing a concrete example, certain embodiments of the disclosure described herein comprise this kind of generally elongate two-part system employing refillable cartridges. However, it will be appreciated the underlying principles described herein may equally be adopted for other electronic cigarette configurations, for example modular systems comprising more than two parts, as devices conforming to other overall shapes, for up example based on so-called box-mod high performance devices that typically have a more boxy shape, or even systems comprising one part where the aerosol provision device and article are integrally formed with one another.

[0029] FIG. 1 is a highly schematic diagram (not to scale) of an example aerosol provision system **10**, such as an e-cigarette, to which embodiments are applicable. The aerosol provision system **10** has a generally cylindrical shape, extending along a longitudinal or y axis as indicated by the axes (although aspects of the invention are applicable to e-cigarettes configured in other shapes and arrangements), and comprises two main components, namely an aerosol provision device **20** and an article **30**.

[0030] The article **30** comprises or consists of aerosol-generating material **32**, part or all of which is intended to be consumed during use by a user. An article **30** may comprise one or more other components, such as an aerosol-generating material storage area **39**, an aerosol-generating material transfer component **37**, an aerosol generation area, a housing, a wrapper, a mouthpiece **35**, a filter and/or an aerosol-modifying agent.

[0031] An article **30** may also comprise an aerosol generator **36**, such as a heating element, that emits heat to cause the aerosol-generating material **32** to generate aerosol in use. The aerosol generator **36** may, for example, comprise combustible material, a material heatable by electrical conduction, or a susceptor. It should be noted that it is possible for the aerosol generator **36** to be part of the aerosol provision device **20** and the article **30** then may comprise the aerosol-generating material storage area **39** for the aerosol-generating material **32** such that, when the article **30** is coupled with the aerosol provision device **20**, the aerosol-generating material **32** can be transferred to the aerosol generator **36** in the aerosol provision device **20**. It should be appreciated that the aerosol generator **36** may encompass an aerosol generator other than a heater. More generally, an aerosol generator is an apparatus configured to cause aerosol to be generated from the aerosol-generating material. In some other embodiments, the aerosol generator is configured to cause an aerosol to be generated from the aerosol-generating material without heating. For example, the aerosol generator may be configured to subject the aerosol-generating material to one or more of vibration, increased pressure, or electrostatic energy.

[0032] Aerosol-generating material is a material that is capable of generating aerosol, for example when heated, irradiated or energized in any other way. The aerosol-generating material **32** may, for example, be in the form of a solid, liquid or gel which may or may not contain an active substance and/or flavourants. In some embodiments, the aerosol-generating material **32** may comprise an “amorphous solid”, which may alternatively be referred to as a “monolithic solid” (i.e. non-fibrous). In some embodiments, the amorphous solid may be a dried gel. The amorphous solid is a solid material that may retain some fluid, **15** such as liquid, within it. In some embodiments,

the aerosol-generating material **32** may for example comprise from about 50 wt %, 60 wt% or 70 wt % of amorphous solid, to about 90 wt %, 95 wt% or 100 wt % of amorphous solid.

[0033] The aerosol-generating material comprises one or more ingredients, such as one or more active substances and/or flavourants, one or more aerosol-former materials, and optionally one or more other functional materials such as pH regulators, colouring agents, preservatives, binders, fillers, stabilizers, and/or antioxidants.

[0034] The active substance as used herein may be a physiologically active material, which is a material intended to achieve or enhance a physiological response. The active substance may for example be selected from nutraceuticals, nootropics, and psychoactives.

[0035] The active substance may be naturally occurring or synthetically obtained. The active substance may comprise for example nicotine, caffeine, taurine, theine, vitamins such as B6 or B12 or C, melatonin, cannabinoids, or constituents, derivatives, or combinations thereof. The active substance may comprise one or more constituents, derivatives or extracts of tobacco, cannabis or another botanical.

[0036] In some embodiments, the active substance comprises nicotine. In some embodiments, the active substance comprises caffeine, melatonin or vitamin B12.

[0037] The aerosol provision device **20** includes a power source **14**, such as a battery, configured to supply electrical power to the aerosol generator **36**. The power source **14** in this example is rechargeable and may be of a conventional type, for example of the kind normally used in electronic cigarettes and other applications requiring provision of relatively high currents over relatively short periods. The power source **14** may be recharged through the charging port (not illustrated), which may, for example, comprise a USB connector.

[0038] The aerosol provision device **20** includes device control circuitry **28** configured to control the operation of the aerosol provision system **10** and provide conventional operating functions in line with the established techniques for controlling aerosol provision systems such as electronic cigarettes. The device control circuitry (processor circuitry) **28** may be considered to logically comprise various sub-units/circuitry elements associated with different aspects of the electronic cigarette's operation. For example, depending on the functionality provided in different implementations, the (device) control circuitry **28** may comprise power source control circuitry for controlling the supply of electrical power from the power source **14** to the aerosol generator **36**, user programming circuitry for establishing configuration settings (e.g. user-defined power settings) in response to user input, as well as other functional units/circuitry associated functionality in accordance with the principles described herein and conventional operating aspects of electronic cigarettes. It will be appreciated the functionality of the (device) control circuitry **28** can be provided in various different ways, for example using one or more suitably programmed programmable computer(s) and/or one or more suitably configured application-specific integrated circuit(s)/circuitry/chip(s)/chipset(s) configured to provide the desired functionality.

[0039] The aerosol provision device **20** has an interface configured to receive the article **30**, thereby facilitating the coupling between the aerosol provision device **20** and the article **30**. The interface is located on a surface of the aerosol provision device **20**.

[0040] The housing of the article **30** has a surface configured to be received by the interface on the aerosol provision device **20** in order to facilitate coupling between the article **30** and the aerosol provision device **20**. The surface of the article may be configured to be a size and/or shape that mirrors the size and/or shape of the interface in order to facilitate coupling between the aerosol provision device **20** and the article **30**. For example, the interface may comprise a cavity, chamber or other space on the surface of the aerosol provision device **20**. The surface of the article **30** can then be configured to be a size and shape that mirrors the size and shape of the cavity in order for the surface of the article **30** to be inserted into the cavity.

[0041] Although not illustrated, the interface of the aerosol provision device **20** and the surface of the article **30** may have complementary features to reversibly attach and mate the article **30** to the

aerosol provision device **20**, such as a screw thread, bayonet fitting, latched or friction fit fixing or other fastening means.

[0042] The interface also comprises one or more connectors, such as contact electrodes, connected via electrical wiring to the control circuitry **28** and the power source **14**. The article **30** also comprises one or more connectors, such as contact electrodes, connected via electrical wiring to the aerosol generator **36**. In use, the article **30** is received by the interface of the aerosol provision device **20**, thereby coupling the aerosol provision device **20** and the article **30**. This results in the connectors on the article **30** mating with the connectors on the aerosol provision device **20**, thereby allowing electrical power and electrical current to be supplied from the power source **14** of the aerosol provision device **20** to the aerosol generator **36** of the article **30**.

[0043] The housing of the article **30** has a surface configured to engage with an interface on the aerosol provision device **20** in order to facilitate coupling between the article **30** and the aerosol provision device **20**. In other words, the aerosol provision device **20** is configured to receive the article **30**, via the interface, and the surface of the article is proximate to the interface on the aerosol provision device **20** when the article **30** is received by the interface.

[0044] The aerosol provision system **10** includes one or more air inlets **21**, located on one or more of the aerosol provision device **20** and the article **30**. In use, as a user inhales on the mouthpiece **35**, air is drawn into the aerosol provision system **10** through the air inlets **21** and along an air channel **23** to the aerosol generator **36**, where the air mixes with the vaporised aerosol-generating material **32** and forms a condensation aerosol. The air drawn through the aerosol generator **36** continues along the air channel **23** to a mouthpiece **35**, carrying some of the aerosol with it, and out through the mouthpiece **35** for inhalation by the user.

[0045] By way of a concrete example, the article **30** comprises a housing (formed, e.g., from a plastics material), an aerosol-generating material storage area **39** formed within the housing for containing the aerosol-generating material **32** (which in this example may be a liquid which may or may not contain nicotine), an aerosol-generating material transfer component **37** (which in this example is a wick formed of e.g., glass or cotton fibres, or a ceramic material configured to transport the liquid from the reservoir using capillary action), an aerosol-generating area containing the aerosol generator **36**, and a mouthpiece **35**.

[0046] Although not shown, a filter and/or aerosol modifying agent (such as a flavour imparting material) may be located in, or in proximity to, the mouthpiece **35**. The aerosol generator **36** of this example comprises a heater element formed from an electrically resistive material (such as NiCr8020) spirally wrapped around the aerosol-generating material transfer component **37**, and located in an air channel **23**. The area around the heating element and wick combination is the aerosol-generating area of the article **30**.

[0047] As illustrated in FIG. **1**, the aerosol provision system **10** also comprises a haptic component **22**. The haptic component **22** is configured to generate one or more vibrations and/or forces in order to provide a haptic sensation to the user of the aerosol provision system **10**. The haptic component **22** can comprise any suitable component configured to provide a haptic sensation (feedback) to a user. For example, the haptic component **22** may comprise an eccentric rotating mass (ERM) or linear resonant actuator (LRA) haptic component. The haptic component **22** may comprise a piezoelectric actuator configured to produce one or more vibrations, and/or the haptic component **22** can comprise a motor configured to produce one or more forces. For example, a pneumatic motor, valve and/or fan can be used to generate a puff, pulse or other movement of air that may be sensed by the user of the aerosol provision system **10**. Additionally, or alternatively, the haptic component **22** may generate haptic sensations that do not directly result from a motion (e.g., vibration) of a mass or of air; e.g., the haptic component **22** may be configured to generate a haptic sensation via an electromagnetic field or via an electric pulse, for example

[0048] The haptic component **22** can be located on or within the aerosol provision device **20** and/or the article **30**. For example, the haptic component **22** can be located proximate to or forming part of

the housing of the aerosol provision device **20** such that the vibrations and/or forces generated by the haptic component **22** can be felt by the user of the aerosol provision system **10** when the user of the aerosol provision system **10** is holding the aerosol provision device **20**. The aerosol provision system **10** may comprise more than one haptic component **22**, for example one located proximate to or forming part of the housing of the aerosol provision device **20** and one located proximate to or forming part of the article **30**, such as on the mouthpiece **35** such that the vibrations/forces generated by the haptic component **22** are sensed on the lips of the user during an inhalation.

[0049] The haptic component **22** can be operatively coupled to the power source **14** of the aerosol provision device **20** in order to receive electrical power, and the haptic component **22** can be operatively coupled to the control circuitry **28** such that the control circuitry **28** can be configured to control the haptic component **22**.

[0050] The aerosol provision system **10** illustrated in FIG. **1** also comprises an orientation sensor **24**. The orientation sensor **24** is configured to detect or measure an orientation of the aerosol provision system **10**. For example, the orientation sensor **24** can comprise a gyroscope, such as a microelectromechanical systems (MEMS) gyroscope or gyrometer, and/or an Inertial Measurement Unit (IMU). The orientation sensor **24** can be located on or within the aerosol provision device **20** and/or the article **30**. The orientation sensor **24** is operatively coupled to the power source **14** of the aerosol provision device **20** in order to receive electrical power, and the orientation sensor **24** is operatively coupled to the control circuitry **28** such that the control circuitry **28** can be configured to control the orientation sensor **24**.

[0051] The control circuitry **28** is configured to determine an orientation of the aerosol provision system **10** based on one or more readings from the orientation sensor **24**. In other words, the control circuit **28** is configured to read or otherwise receive readings from the orientation sensor **24**, and use the readings to determine the orientation of the aerosol provision system **10**, such as substantially vertically (along the y-axis in FIG. **1**), substantially horizontally (along the x-axis in FIG. **1**) or another orientation. The control circuitry **28** can receive readings from the orientation sensor **24** periodically, for example every second, minute or five minutes, or the control circuitry **28** can send a request to the orientation sensor **24** for the one or more readings. Equally, the control circuitry **28** can be configured to determine the orientation of the aerosol provision system **10** in response to an event, for example one or more readings from the orientation sensor **24** changing, thereby indicating a change in orientation of the aerosol provision system **10**.

[0052] In response to determining the orientation of aerosol provision system **10**, the control circuitry **28** is configured to adjust the setting of the haptic component **22**. The setting can be one or more of a duration, magnitude, amplitude, frequency and/or pattern of the one or more vibrations or forces generated by the haptic component. For example, the setting can be an amplitude and/or frequency of a vibration generated by the haptic component. Equally, the setting can be a duration and/or magnitude of a force generated by the haptic component. Adjusting the setting of the haptic component **22** can also correspond to disabling the haptic component **22**. In other words, in response to determining the orientation of the aerosol provision system **10**, the control circuitry **28** can be configured to disable the haptic component **22**. For example, the control circuitry **28** can be configured to prevent the supply of electrical power from the power source **14** to the haptic component **22**. Equally, adjusting the setting of the haptic component **22** can also correspond to enabling the haptic component **22**, for example by enabling the supply of electrical power from the power source **14** to haptic component **22**.

[0053] Adjusting a setting of the haptic component **22** by the control circuit **28** can be dependent on the determined orientation. For example, an amplitude of the vibration generated by the haptic component may be altered in response to determining that the aerosol provision system **10** is in a downward orientation (i.e. with the mouthpiece **35** being at the lowest point on the aerosol provision system **10**, in opposite orientation in the y-direction to that illustrated in FIG. **1**) whilst a frequency may be altered in response to determining that the aerosol provision system **10** is

oriented substantially horizontal (i.e. along the x-axis in FIG. 1).

[0054] Although not illustrated in FIG. 1, the aerosol provision system **10** can also comprise a movement sensor. The movement sensor is configured to detect or measure a movement or motion of the aerosol provision system **10**. For example, the movement sensor can be an accelerometer and/or an IMU. Where the movement sensor is an IMU, the orientation sensor **24** and the movement sensor can be incorporated into a single IMU. In other words, the aerosol provision system **10** comprises an IMU incorporating the orientation sensor **24** and the movement sensor. The movement sensor can be located on or within the aerosol provision device **20** and/or the article **30**. The movement sensor is operatively coupled to the power source **14** of the aerosol provision device **20** in order to receive electrical power, and the movement sensor is operatively coupled to the control circuitry **28** such that the control circuitry **28** can be configured to control the movement sensor.

[0055] The control circuitry **28** can be further configured to determine a movement of the aerosol provision system **10** based on one or more readings from the movement sensor. In other words, the control circuitry **28** is configured to read or otherwise receive readings from the movement sensor, and use the readings to determine a movement of the aerosol provision system **10**. In a similar fashion to as described above in respect of the orientation sensor **24**, the control circuitry **28** can receive readings from the movement sensor periodically, for example every second, minute or five minutes, or the control circuitry **28** can send a request to the movement sensor for the one or more readings. Equally, the control circuitry **28** can be configured to determine a movement of the aerosol provision system **10** in response to an event, for example one or more readings from the movement sensor changing, thereby indicating an acceleration or other motion of the aerosol provision system **10**. The event for determining an orientation of the aerosol provision system **10** can be the determination of the movement of the aerosol provision system **10**. For example, in response to determining a movement of aerosol provision system **10**, the control circuitry **28** can be configured to determine an orientation of the aerosol provision system **10**. In other words, having determined that the aerosol provision system **10** is moved, the control circuitry **28** can be configured to determine the new orientation of the aerosol provision system **10** based on one or more readings from the orientation sensor **24**.

[0056] In response to determining a movement or change of the orientation of aerosol provision system **10**, the control circuitry **28** is configured to adjust the setting of the haptic component **22**. The setting can be one or more of a duration, magnitude, amplitude, frequency and/or pattern of the one or more vibrations or forces generated by the haptic component **22**. Equally, as described above, adjusting the setting can correspond to disabling or enabling the haptic component **22**. The setting adjusted by the control circuitry **28** in response to determining a movement or change of the orientation of aerosol provision system **10** may be the same as or different to the setting adjusted based on the determined orientation. For example, the frequency of the vibrations/forces generated by the haptic component **22** may be increased in response to determining a movement of the aerosol provision system **10** and in response to determine an orientation of the aerosol provision system **10**, or the amplitude of the vibrations/forces generated by the haptic component **22** may be increased in response to determining an orientation of the aerosol provision system **10** whilst the haptic component **22** may be disabled in response to determining a movement of the aerosol provision system **10**.

[0057] Adjusting a setting of the haptic component **22** by the control circuitry **28** can be dependent on the determined movement and/or orientation of the aerosol provision system **10**. For example, an amplitude of the vibration/force generated by the haptic component **22** may be altered in response to determining that the aerosol provision system **10** falling downwards whilst a frequency of the vibration/force generated by the haptic component may be altered in response to determining that the aerosol provision system **10** is being rotated upwards (e.g. a movement similar to a user moving the aerosol provision system **10** from their waist up to their lips to take an inhalation on the

aerosol provision system **10**). Equally, the haptic component **22** may be disabled in response to determining that the aerosol provision system **10** is not moving (i.e. stationary with no movement detected) and the haptic component **22** may be enabled in response to determining a movement of the aerosol provision system **10**.

[0058] As discussed, the setting(s) of the haptic component **22** may be adjusted according to the orientation and/or movement of the aerosol provision system **10**. Accordingly, in some implementations, when the haptic component **22** is controlled to provide an output, e.g., a vibration, the output of the haptic component **22** may also vary in accordance with the variation in the setting. That is to say, for example, as the orientation of the aerosol provision system **10** changes, i.e., moves from a vertical position in which the mouthpiece points upwards to a vertical position in which the mouthpiece points downwards, the output (e.g., amplitude) of the haptic component **22** changes with the orientation. Depending on the frequency at which the orientation and/or movement sensor determines the orientation or movement of the aerosol provision system, the setting (and thus the output) of the haptic component **22** may be varied at multiple times during a particular movement/change in orientation. For example, if one assumes that the aerosol provision system **10** at a vertical position in which the mouthpiece points upwards is a 0° position and a vertical position in which the mouthpiece points downwards is a 180° position, the output (e.g., amplitude) of the haptic component **22** may be set to a first level when the aerosol provision system **10** is at the 0° position, to a second level when the aerosol provision system **10** is at a 90° position, and a third level when the aerosol provision system **10** is at the 180° position. In this way, it should be understood that certain patterns of haptic feedback can be provided to the user based on the change of orientation or the transition between different orientations.

[0059] In some implementations, the ability for the haptic component **22** to vary its settings and/or output in accordance with the orientation or movement of the aerosol provision system **10** may be selectable. That is, a user may be able to switch on the ability of the haptic component **22** to vary its settings and/or output in accordance with the orientation or movement of the aerosol provision system **10**, e.g., via a button press or other input on the aerosol provision device or via a remote device (such as a smartphone or the like) communicatively coupled to the aerosol provision system **10**. In this way, the user may be provided with the option to selectively turn on and off the ability of the haptic component **22** to vary its settings and/or output in accordance with the orientation or movement of the aerosol provision system **10** may be selectable.

[0060] Although not illustrated in FIG. 1, the aerosol provision system **10** may comprise one or more output components, such as a display component and/or a speaker component. The control circuitry **28** can be configured to adjust a setting of the display component and/or the speaker component based on at least one of the determined orientation and the determined movement. In other words, a setting of the one or more output components can be adjusted in response to determining the orientation and/or movement aerosol provision system **10**.

[0061] In the case of the display component, the setting of the display component can correspond to a brightness, contrast, colour balance or other visual property of the display component (i.e. the screen of the display component). In the case of the speaker component, the setting of the speaker component can correspond to a volume, balance or other auditory property of the speaker component. Equally, adjusting the setting of the display device and/or speaker component can correspond, respectively, to enabling or disabling the display component and/or speaker component. Adjusting the setting of the display component and/or the speaker component could occur substantially simultaneously with adjusting the setting of the haptic component **22**, or in response to adjusting the setting of the haptic component **22**. For example, the control circuitry **28** could be configured to adjust the brightness of the display component (e.g. lower the brightness) and/or adjust the volume of the speaker component (e.g. lower the volume) in response to enabling the haptic component **22**. Equally, the control circuitry **28** could be configured to disable the display component and/or the speaker component when the haptic component **22** is enabled, then

enable the display component and/or the speaker component when the haptic component **22** is disabled. This allows a primary component for conveying signals to the user to be changed between the one or more output components and the haptic component **22**.

[0062] The aerosol provision system **10** can also comprise a communications interface, and the control circuitry **28** configured to communicate with one or more external devices, such as a computer, mobile phone or other electronic device, via the communications interface. The communications interface can be configured to communicate using a suitable wireless communications protocol such as Wi-Fi, Bluetooth, RFID, NFC. The control circuitry **28** can be configured to adjust a setting of the haptic component **22**, such as the settings described above, based on signal received via the communications interface from an external device. In other words, the external device can send a signal to the aerosol provision system **10** by the communications interface, and in response to receiving the signal, the control circuit **28** adjusts a setting of the haptic component **22**. This allows the setting of the haptic component **22** to be adjusted without directly interacting with the aerosol provision system **10**. The signal received from the external device via the communications interface may indicate the setting and/or magnitude or type of adjustment to be made to the haptic component **22**, or the control circuitry **28** may be configured to determine the setting of haptic component **22** to adjust based on the signal received from the external device.

[0063] The aerosol provision system **10** can also comprise an input device, such as a button, switch, or touchscreen display. In the latter case, the display component described above can be a touchscreen component that is also configured to receive touch inputs. The control circuitry can be configured to adjust a setting of the haptic component **22**, such as the settings described above, based on an input received via the input device. In other words, the user can provide an input on the input device, and in response to receiving the input, the control circuit **28** adjusts a setting of the haptic component **22**. This allows the setting of the haptic component **22** to be adjusted by the user providing an input on the aerosol provision system **10**. The input received from the input device (e.g. from the user) may indicate the setting of the haptic component **22** to be adjusted, or the control circuitry **28** may be configured to determine the setting of haptic component **22** to adjust based on the input received from the input device.

[0064] FIG. **2** is a flow diagram of a method **200** for operating an aerosol provision system, such as aerosol provision system **10**. The method begins at step **210**, where an orientation of the aerosol provision system is determined based on one or more readings from an orientation sensor of the aerosol provision system. At step **220**, a setting of a haptic component of the aerosol provision system is adjusted based on the determined orientation. The method then ends.

[0065] The method **200** illustrated in FIG. **2** may be stored as instructions on a computer readable storage medium, such that when the instructions are executed by a processor, the method described above is performed. The computer readable storage medium may be non-transitory. In other words, the method **200** illustrated in FIG. **2** may be computer implemented. The method **200** may be performed by the aerosol provision device **20**, such as by the control circuitry **28**.

[0066] As described above, the present disclosure relates to (but it not limited to) an aerosol provision system comprising a haptic component, an orientation sensor and control circuitry configured. The control circuitry is configured to determine an orientation of the aerosol provision system based on one or more readings from the orientation sensor, and adjust a setting of the haptic component based on the determined orientation.

[0067] Thus, there has been described an aerosol provision system and method.

[0068] The various embodiments described herein are presented only to assist in understanding and teaching the claimed features. These embodiments are provided as a representative sample of embodiments only, and are not exhaustive and/or exclusive. It is to be understood that advantages, embodiments, examples, functions, features, structures, and/or other aspects described herein are not to be considered limitations on the scope of the invention as defined by the claims or

limitations on equivalents to the claims, and that other embodiments may be utilised and modifications may be made without departing from the scope of the claimed invention. Various embodiments of the invention may suitably comprise, consist of, or consist essentially of, appropriate combinations of the disclosed elements, components, features, parts, steps, means, etc., other than those specifically described herein. In addition, this disclosure may include other inventions not presently claimed, but which may be claimed in future.

Claims

1. An aerosol provision system comprising: a haptic component; an orientation sensor; and control circuitry configured to: determine an orientation of the aerosol provision system based on one or more readings from the orientation sensor; and adjust a setting of the haptic component based on the determined orientation.
2. The aerosol provision system of claim 1, wherein the orientation sensor is a gyroscope.
3. The aerosol provision system of claim 1, further comprising a movement sensor, wherein the control circuitry is further configured to: determine a movement of the aerosol provision system based on one or more readings from the movement sensor; and adjust a setting of the haptic component based on the determined movement.
4. The aerosol provision system of claim 3, wherein the movement sensor is an accelerometer.
5. The aerosol provision system of claim 3, further comprising an Inertial Measurement Unit, IMU, incorporating the orientation sensor and the movement sensor.
6. The aerosol provision system of claim 3, further comprising a display component, and the control circuitry is further configured to adjust a setting of the display component based on the determined movement.
7. The aerosol provision system of claim 3, further comprising a speaker component, and the control circuitry is further configured to adjust a setting of the speaker component based on the determined movement.
8. The aerosol provision system of claim 1, wherein adjusting the setting of the haptic component corresponds to disabling the haptic component.
9. The aerosol provision system of claim 1, wherein the setting is an amplitude and/or frequency of a vibration generated by the haptic component.
10. The aerosol provision system of claim 1, wherein the setting is a magnitude of a force generated by the haptic component.
11. The aerosol provision system of claim 1, further comprising a display component, and the control circuitry is further configured to adjust a setting of the display component based on the determined orientation.
12. The aerosol provision system of claim 6, wherein the setting is a brightness of the display component.
13. The aerosol provision system of claim 6, wherein adjusting the setting of the display component corresponds to disabling the display component.
14. The aerosol provision system of claim 1, further comprising a speaker component, and the control circuitry is further configured to adjust a setting of the speaker component based on the determined orientation.
15. The aerosol provision system of claim 7, wherein the setting is a volume of the speaker component.
16. The aerosol provision system of claim 7, wherein adjusting the setting of the speaker component corresponds to disabling the speaker component.
17. The aerosol provision system of claim 1, further comprising a communications interface, and wherein control circuitry is further configured to adjust a setting of the haptic component based on a signal received via the communications interface from an external device.

- 18.** The aerosol provision system of claim 1, further comprising an input device, and wherein control circuitry is further configured to adjust a setting of the haptic component based on an input received via the input device.
- 19.** An aerosol provision device for an aerosol provision system, the aerosol provision system comprising a haptic component and an orientation sensor, wherein the aerosol provision device comprises control circuitry configured to: determine an orientation of the aerosol provision system based on one or more readings from the orientation sensor; and adjust a setting of the haptic component based on the determined orientation.
- 20.** A method for operating an aerosol provision system comprising: determine an orientation of the aerosol provision system based on one or more readings from an orientation sensor of the aerosol provision system; and adjust a setting of a haptic component of the aerosol provision system based on the determined orientation.
- 21.** (canceled)
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